KINETICS OF RIGID BODIES

Plane Rigidbody motion:

A rigidbody is said to be in motion, if the position of any particle of the body or the orientation of any line drawn on the body is changed during a finite time interval.

Three types of plane motion:

1. Translation
2. Rotation about a fixed axis
3. General plane motion

1. Translation:

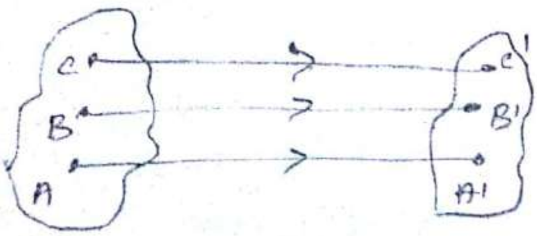
A rigid body is said to be in translation, if the linear displacement of every point in the rigidbody is the same. ie, the orientation of any line drawn on the body remains unchanged, but its position is changed.

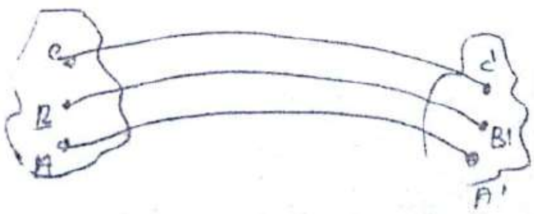
(a) Rectilinear translation

If all paths are parallel in straight line, the motion is said to be rectilinear translation

b) Curvilinear translation

If the paths are congruent curves, the motion is said to be curvilinear translation





2. Rotation about a fixed axis / Axis of rotation:

A rigid body is said to be in rotation, if all the particles of the body move along circle centred on a fixed axis called axis of rotation. The planes of the circular paths are perpendicular to the axis of rotation.

3. General plane motion:

A Rigidbody is said tobe in general plane motion, if it undergoes a combination of translation and rotation. It is neither a pure translation nor a pure rotation. All the particles on the body translate through some distance and also rotate through a certain angle.

1. Angular Displacement: (θ)

Total angle through which a body has rotated

$\theta = f(t)$, It varies with time.

2. Angular Velocity (ω)

It is a rate of change of angular displacement of the body.

$$\omega = \frac{d\theta}{dt} \quad \text{(or)} \quad \omega = \frac{2\pi N}{60}$$

3. Angular acceleration: (α)

It is the rate of change of angular velocity. It is expressed in rad/s²

$$\alpha = \frac{d\omega}{dt}$$

Scanned by CamScanner

Equation of motion under rotation

(a)

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\left.\begin{array}{l}\text{Displacement in} \\ n^{th} \text{ second}\end{array}\right\} = \theta^{nth} = \omega_0 + \frac{\alpha}{2}(2n-1)$$

Relation between Angular motion and Linear motion

(b)

1. Angular displacement = $\dfrac{\text{Linear displacement}}{\text{Radius}}$

2. Angular velocity = $\dfrac{\text{Linear velocity}}{\text{Radius}}$

3. Angular Acceleration = $\dfrac{\text{Linear Acceleration}}{\text{Radius}}$.

1. A flywheel starts rotating from rest and is given an acceleration of $2 \, rad/s^2$

(i) find the angular velocity and speed in rpm after

(ii) If the flywheel is brought to rest with a Uniform retardation of $1.25 \, rad/s^2$, determine the time taken by the flywheel in seconds to come to rest.

G.D

$\omega_0 = 0$, $\alpha = 2 \, rad/s^2$, $t = 60 \, sec$

$\omega = 0$, $\alpha = -1.25 \, rad/s^2$

To find ω, $N = ?$

$t = ?$

$\omega = \omega_0 + \alpha t$

$= 0 + (2 \times 60)$

$\boxed{\omega = 120 \, rad/s}$

$\omega = \frac{2\pi N}{60}$

$N = \frac{\omega \times 60}{2\pi} = \frac{60 \times 120}{2\pi}$

$\boxed{N = 1145.9 \, rpm}$

(ii) $\omega = 0$ (brought to rest) , $\omega_0 = 120 \, rad/s$

$\alpha = -1.25 \, rad/s^2$

$\omega = \omega_0 + \alpha t$

$0 = 120 - 1.25t$

$\boxed{t = 96 \, sec}$

2. A rigid body rotates about a fixed axis and slows down from 300 rpm to 150 rpm in 2 min?

(i) Determine the angular acceleration?

(ii) Determine the number of revolution completed in 2 min?

G.D

$N_0 = 300 rpm$

$\omega_0 = \frac{2\pi N_0}{60}$

$= \frac{2 \times \pi \times 300}{60}$

$\boxed{\omega_0 = 31.42 \, rad/s}$

$N = 150 rpm$

$\omega = \frac{2\pi N}{60}$

$= \frac{2 \times \pi \times 150}{60}$

$\boxed{\omega = 15.71 \, rad/s}$

Scanned by CamScanner

$t = 2\,min = 2 \times 60 = 120\,sec$

$\omega = \omega_0 + \alpha t$

$15.71 = 31.42 + (\alpha \times 120)$

$\boxed{\alpha = -0.1309\ rad/s^2}$

−ve sign indicates angular retardation

$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

$= (31.42 \times 120) + \frac{1}{2}(-0.1307 \times 120^2)$

$= 2827.43\ rad.$

No. of revolution $= \frac{2827.43}{2\pi} = 450$ //
$(n = \theta/2\pi)$

8. The rotation of a fly wheel is governed by the equation $\omega = 3t^2 - 2t + 2$ where ω is in radians per second and t is in seconds. After one second from the start, the angular displacement was 4 radian. Determine the angular displacement, angular velocity, and angular acceleration of the fly wheel when $t = 3$ seconds.

<u>G.D</u>

$\omega = 3t^2 - 2t + 2$

$\theta = 4\ radian.$

$\omega_0 = 0, \quad t = 1\,sec.$

To find

$\theta = ?$

$\omega = ?$

$\alpha = ?$

<u>~~θz~~</u>

~~$\alpha = \frac{d\omega}{dt}$~~

~~$\boxed{\alpha = 6t - 2}$~~

$$\theta = \int d\omega \, dt$$

$$\boxed{\theta = \frac{3t^3}{3} - \frac{2t^2}{2} + 2t + c} \longrightarrow (1)$$

$$4 = \frac{3(1)^3}{2} - \frac{2(1)^2}{2} + 2(1) + c$$

$$\boxed{c = 5.5}$$

sub c in (1)

$$\boxed{\theta = \frac{3t^3}{3} - \frac{2t^2}{2} + 2t + 5.5}$$

$$\boxed{\omega = \frac{9t^2}{2} - \frac{4t}{2} + 2} \Rightarrow \omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt} = \frac{18t}{2} - \frac{4}{2}$$

$$\boxed{\alpha = 9t - 2}$$

sub $t = 3$ sec in θ, ω, α eqn

$$\theta = \frac{3(3)^3}{3} - \frac{2(3)^2}{2} + 2(3) + 5.5$$

$$\boxed{\theta = 29.5 \text{ radian}}$$

$$\omega = \frac{3(3)^2}{2} - \frac{\cancel{4}^{2}(3)}{\cancel{2}} + 2$$

$$\boxed{\omega = 9.8 \text{ rad/s}}$$

$$\alpha = 9(3) - 2$$

$$\boxed{\alpha = 25 \text{ rad/s}^2}$$

A grinding wheel is attached to a shaft of an electric motor having a rated speed 3600 rpm. When the power is turned on the unit reaches its maximum rated speed in 5 second and when the power is turned off the unit comes to rest in 70 s. Assuming uniformly accelerated motion, Determine the number of revolution made by the wheel

(a) in reaching its rated speed (b) in coming to rest.

G.D

Max speed $\xrightarrow{\text{comes to}}$ Rest

$N = 3600\,rpm$

$t = 5\,sec$

$t = 70\,sec$

$\omega = 0$

To find

n at max. speed $= ?$

n at which unit comes to rest $= ?$

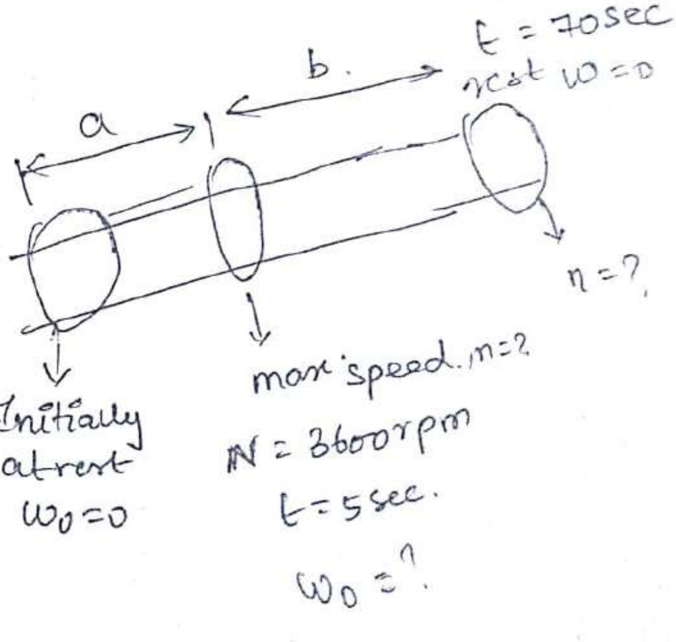
Formula:

$$n = \theta / 2\pi$$

Soln

portion a

$$\omega = \frac{2\pi N}{60} = \frac{2 \times \pi \times 3600}{60}$$



$$\boxed{\omega = 276.99\,rad/s}$$

$$\omega = \omega_0 + \alpha t$$

$$276.99 = 0 + \alpha(5)$$

$$\boxed{\alpha = 55.398\,rad/s^2}$$

$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

$\theta = 0(5) + \frac{1}{2} (55.398 \times 5^2)$

$\boxed{\theta = 692.475 \text{ rad}}$

at $t = 5$ sec $\quad n = \theta/2\pi = \dfrac{692.475}{2 \times \pi}$

No. of revolution $\boxed{n = 110.}$

(ii) When power is switched off.

$t = 70$ sec.

$\omega_0 = 276.998$ rad/s

$\omega = 0$ rad/s

$\omega = \omega_0 + \alpha t$

$0 = 276.998 + \alpha (70)$

$\boxed{\alpha = -3.957 \text{ rad/s}^2}$

$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

$= (276.99 \times 70) + \frac{1}{2} (-3.957) \times 70^2$

$\theta = 9694.65$ rad

$n = \theta/2\pi = \dfrac{9694.65}{2\pi}$

$\boxed{n = 1543}$

During the starting phase of a Computer, it is observed that a storage disk which is initially at rest, executed 2.5 revolution in 0.3 seconds. Assume that the motion was uniformly acceleration. Determine a) The angular acceleration of the disk

(b) The final angular velocity of the disk.

G.D

Initially at rest $\omega_0 = 0$.

$n = 2.5$

$t = 0.3$ second.

To find

$\alpha = ?$

$\omega = ?$

Soln

$n = \theta / 2\pi$

$\theta = 2.5 \times 2 \times \pi$

$$\boxed{\theta = 15.71 \text{ rad}}$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$$

$$15.71 = 0 + \frac{1}{2} \times \alpha \times 0.3^2$$

$$\boxed{\alpha = 349.11 \text{ rad/s}^2}$$

$$\omega = \omega_0 + \alpha t$$

$$\omega = 0 + 349.11 \times 0.3$$

$$\boxed{\omega = 104.73 \text{ rad/sec}}$$

$(u_c)_0 = r\omega_0$ r = radius of pulle

(a) Load A is connected to a double pulley by one of the two inextensible cables shown in fig. The motion of the pulley is controlled by cable B, which has a constant acceleration of 1 m/s² and the initial velocity of 1·5 m/s, both directed to the right. Determine (i) Number of revolution executed by the pulley in 5 seconds

(ii) Velocity and change in position of the load A, after 5 seconds. (S)

<u>G.P</u>

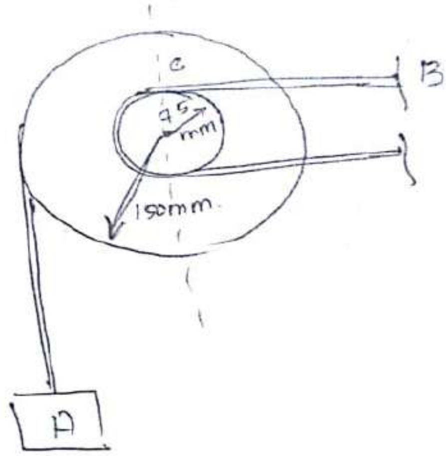
$a = 1 m/s^2$

$u = 1.5 m/s$

<u>To find</u>

n at $t = 5$ second $= ?$

$V = ?$, $S = ?$

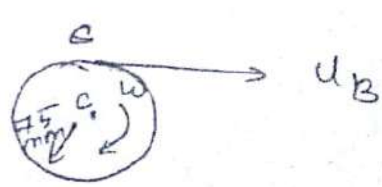


Soln:

(i) No. of. revolution of pulley in 5sec.:

Note: (Velocity at B is acting towards right, hence the inner pulley is rotating in clockwise direction, with an angular velocity of 'ω'. As the cable is inextensible, the velocity of point B is equal to the velocity of point C. and tangential. Component of acceleration of C is equal to the acceleration of C.)

$(u_B)_0 = (u_C)_0 = 1.5 m/s\ (\rightarrow)$



Initial acceleration of B and C,

$(a_C)_0 = (a_B)_0 = 1 m/s^2\ (\rightarrow)$

$V = r\omega$

$(u_c)_0 = r\omega_0$ r = radius of pulley in m

$1.5 = \left(\frac{75}{1000}\right)\omega_0$

(a) $\boxed{\omega_0 = 20\ rad/s}$ (clockwise)

$a = r \times \alpha$

$1 = \alpha\left(\frac{75}{1000}\right)$

$\boxed{\alpha = 13.33\ rad/s^2}$

$\omega = \omega_0 + \alpha t$ $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

(b) At $t = 5$ sec.

$\omega = 20 + (13.33 \times 5)$ $\theta = (20 \times 5) + (\frac{1}{2} \times 13.33 \times 5^2)$

$\boxed{\omega = 86.65\ rad/s}$ $\boxed{\theta = 266.65\ rad}$

(Clockwise)

$n = \theta/2\pi = \frac{266.65}{2\pi} = 42.43$ revolution

(ii) velocity and change in position of load A in 5 sec., $r = 150mm$, $\omega = 20 m/s$, $\theta = 266.65\ rad$

$V = r\omega$ $S = r\theta$

$(u)_A = \left(\frac{150}{1000}\right) 20$ $= \left(\frac{150}{1000}\right) \times 266.65$

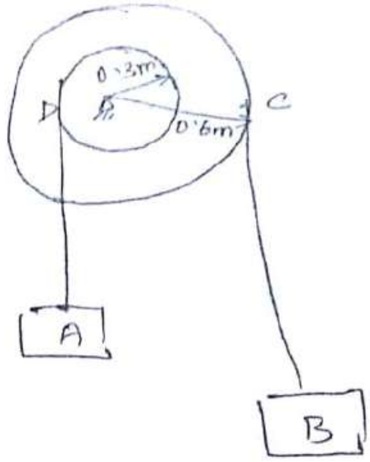
$\boxed{(u)_A = 3 m/s}$ (↑) $\boxed{S = 40m}$ (↑)

S. No. ...

A pulley with two loads, connected by an inextensible cord as shown in fig. If the load B moves downward with an initial velocity of 1.5 m/s and uniform acceleration of 0.75 m/s², determine

(i) Number of revolutions executed by the pulley in 2 sec

(ii) Velocity and position of the load A after 2 sec



G.D

Load B:

$u = 1.5 m/s$

$a = 0.75 m/s^2$

To find

$n_{at\ t = 2sec} = ?$

$v, s_{at\ t = 2sec} = ?$

Soln

Since the cable is inextensible, the velocity and acceleration of C, a point on the rim of the outer pulley = velocity and acceleration of load B

* $(u_B)_0 = 1.5 m/s$, $(a_B)_0 = 0.75 m/s^2$

$r = 0.6m$, Equation $v = r\omega$, $a = r \times \alpha$

* $(u_c)_0 = 1.5 m/s$, $(a_c)_0 = 0.75 m/s^2$

$(\omega_c)_0 = \frac{(u_c)_0}{r} = \frac{1.5}{0.6}$, $(\alpha_c)_0 = \frac{(a_c)_0}{r} = \frac{0.7}{0.6}$

$\boxed{(\omega_c)_0 = 2.5 rad/s}$ $\boxed{(\alpha_c)_0 = 1.25 rad/s^2}$

Number of revolution of the pulley in 2 sec

$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ $(t = 2sec)$

$= (2.5 \times 2) + (\frac{1}{2} \times 1.25 \times 2^2)$

$\boxed{\theta = 7.5 \, rad}$

$n = \theta/2\pi = \frac{7.5}{2\pi} = 1.194$

$\boxed{n = 1 \text{ revolution}}$

(ii) velocity and position of load A after 2 sec.

$(V, a)_D = (V, a)_A$, $(\omega, \alpha)_D = (\omega, \alpha)_A$

$(V, a)_D = (V, a)_C$

$(\omega, \alpha)_C = (\omega, \alpha)_D$

$[(\omega_C)_0 = 2.5 \, rad/s = (\omega_D)_0$

$(\alpha_C)_0 = 1.25 \, rad/s^2 = (\alpha_D)_0$

$t = 2 \, sec]$, $\theta_A = 7.5 \, rad$

$\omega = \omega_0 + \alpha t$

$= 2.5 + 1.25 \times 2$

$\boxed{\omega = 5 \, rad/s}$

$r = 0.3 \, m$, $V_A = r_A \omega_A$

$= 0.3 \times 5$

$\boxed{V_A = 1.5 \, m/s}$

$S_A = r_A \times \theta_A$

$= 0.3 \times 7.5$

$\boxed{S_A = 2.25 \, m}$

Scanned by CamScanner

$$a_c = \sqrt{(a_n)^2 + (a_t)^2}$$

(a) The cylinder as shown in fig. and the thin inextensible string wrapped around it are initially at rest. If the cylinder has a uniform acceleration of 100 rad/s² counterclockwise, determine

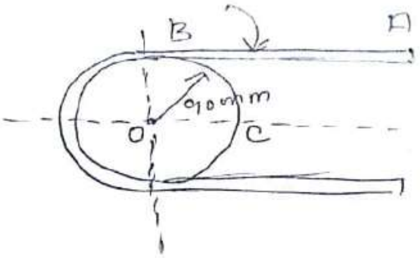
(a) the acceleration of point A of the string

(b) the magnitude of the acceleration of point C of the cylinder after 3.5 seconds.

<u>G.D</u>

$\omega_0 = 0$

$\alpha = 100\, rad/s^2$



<u>To find</u>

$a_A = ?$

a_c at $t = 3.5\,sec = ?$

<u>Soln</u>

(i) 'a' at A

String initially at rest, so $u = 0$

$a_A = r\alpha$

$90\,mm = 90 \times 10^{-3}\,m$

$a_A = \frac{90}{1000}\alpha$

$a_A = 90 \times 10^{-3}(100)$

$\boxed{a_A = 9\,ms^{-2}}$

(ii) Acceleration at 'c' after 3.5 sec

Point C - both angular and linear

Note, linear - Tangential

Angular - Normal + tangential

$$a_c = \sqrt{(a_n)^2 + (a_t)^2}$$

Tangential acceleration

$$\boxed{a_t = 9 ms^{-2}}$$

Normal acceleration

$$a_n = r\omega^2$$

$$\boxed{a_n = (90 \times 10^{-3})\omega^2} \rightarrow$$

$$\omega = \cancel{\omega_0}^{0} + \alpha t$$

$$\omega = 100(3.5)$$

$$\boxed{\omega = 350 \ rad \ s^{-1}}$$

sub ω in (1)

$$a_n = (90 \times 10^{-3})(350)^2$$

$$\boxed{a_n = 11025 \ ms^{-1}}$$

$$a_c = \sqrt{a_n^2 + a_t^2}$$

$$= \sqrt{11025^2} + 9^2$$

$$\boxed{a_c = 11025 \ ms^{-2}}$$

Ans:

$$a_A = 9 \ ms^{-2}$$

$$a_c = 11025 \ ms^{-2}$$

Scanned by CamScanner

D' Alembert Principle

Block

1) If the block is moving in $\boxed{X\text{-direction}}$

(a) Right side direction

$$\Sigma F_x = ma, \ \Sigma F_y = 0$$

(b) Left side direction

$$\Sigma F_x = -ma, \ \Sigma F_y = 0$$

2) If the block is moving in $\boxed{Y\text{-direction}}$

(a) Downward direction

$$\Sigma F_y = -ma, \ \Sigma F_x = 0$$

(b) Upward direction

$$\Sigma F_y = ma, \ \Sigma F_x = 0$$

Pulley

(a) If the pulley is rotating in clockwise direction by the tension rope

$$\Sigma M_G = I_G \times \alpha$$

where

ΣM_G - sum of all moments about the centre of pulley.

I_G - mass moment of inertia of pulley about centre.

α = Angular acceleration $\boxed{\alpha = a/r}$

r = radius, a = linear acceleration.

[* If the radius of pulley is known, $I_G = \frac{m_p r^2}{2}$

* If the radius of gyration (k) is known / given,

$$I_G = m_p k^2$$]

$$[\Sigma M_G = \text{Tension} \times \text{radius}]$$

(b) If the pulley is rotating in anticlockwise direction by the tension rope

$$\Sigma M_G = -\boxed{I_G \times \alpha}$$

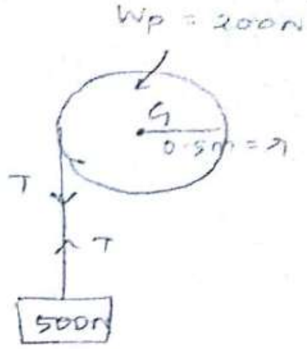
1. 2) ... PRINCIPLE

(a) The ~~reel~~ block of 500N is suspended by a rope around pulley of weight 200N and radius 0.5m as shown. Determine the acceleration of the weight and tension in the rope

<u>G.D</u>

$W_B = 500N$, $W_P = 200N$

$r = 0.5m$



<u>To find</u>

$a = ?$, $T = ?$

<u>Solution</u>

(b) Due to the weight 500N, the pulley rotates in Anticlockwise direction. and block moves towards downward direction.

<u>Block</u>

$$\boxed{\Sigma F_y = -ma}$$

$\Sigma F_x = 0$ (Not used)

<u>pulley</u> D'Alembert

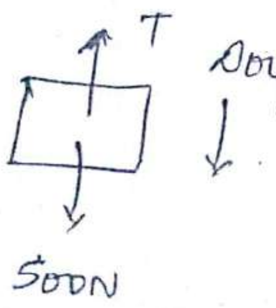
$$\boxed{\Sigma M_G = -I_G \times \alpha}$$

$$I_G = \frac{mr^2}{2}$$

$\Sigma M_G = T \times r$

<u>FBD : Block</u>

$m_B = 500/9.81 = 50.96 kg$



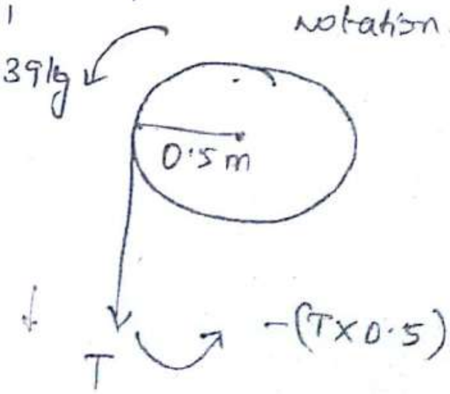
$\Sigma F_y = T - 500$

<u>FBD : pulley</u>

$$m_p = \frac{200}{9.81}$$

$m_p = 20.39 kg$

Anticlockwise rotation.



$\Sigma M_G = -(T \times 0.5)$

$$I_G = \frac{20.39 \times 0.5^2}{2} = 2.548$$

NCTD.

PRINCIPLE

$\Sigma F_y = -ma$

$T - 500 = -50.96 \times a$

$\boxed{T + 50.96a = 500} \rightarrow (2)$

$\alpha = a/r = a/0.5 = 2a$

$\Sigma M = -I_G \times \alpha$

$-(T \times 0.5) = -(2.548 \times 2a)$

$0.5T = (2.548 \times 2)a$

$\boxed{T = 10.192a} \rightarrow (1)$

Sub $T = 10.192a$ in (2)

$T + 50.96a = 500$

$(10.192a) + (50.96a) = 500$

$\boxed{a = 8.176 \, m/s^2}$

$\alpha = a/r = \frac{8.176}{0.5}$

$\boxed{\alpha = 16.35 \, rad/s^2}$

Sub $a = 8.176 \, m/s^2$ in (1)

$T = 10.192a$

$= 10.192 \times 8.176$

$\boxed{T = 83.33 \, N}$

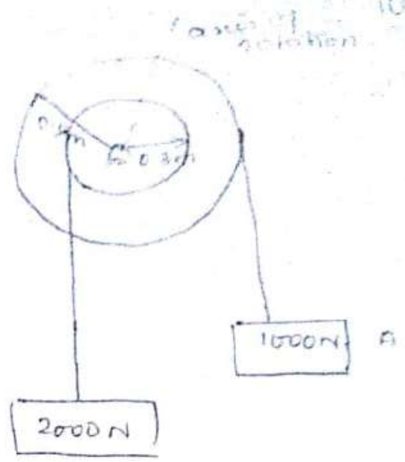
A Composite pulley of weight 500N and radius of gyration of 0.4m is attached with two blocks A and B of weight 1000N and 2000N respectively, as shown in fig. Determine the angular acceleration of the pulley and tension in the string by using D' Alembert principle.

note:

D'ALEMBERT PRINCIPLE

The Inner and outer pulley rotates in anticlockwise direction due to the weight 2000N.

(a)



FBD: B

$W_B = 2000N$

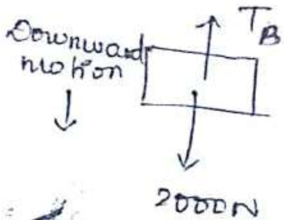
$m_B = 203.87 kg$

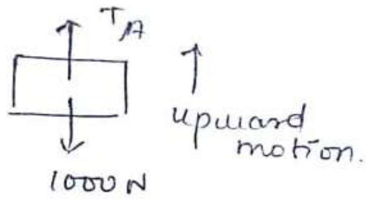
FBD: A

$W_A = 1000 N$

$m_A = 101.936 kg$

(b)





$\Sigma F_y = -ma$

$\Sigma F_y = T_B - 2000N$

$T_B - 2000 = -203.87 \times a$

$T_B = -203.87a + 2000$

$\boxed{T_B = -203.87a + 2000}$

$\Sigma F_y = +ma.$

$\Sigma F_y = T_A - 1000$

$T_A - 1000 = 101.936 \times a.$

$\boxed{T_A = 101.936a + 1000}$

FBD: pulley

$\Sigma M_G = -I_G \times \alpha$

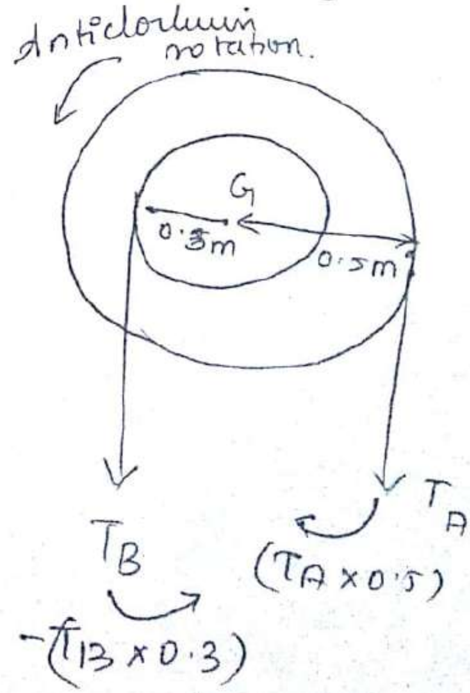
$\Sigma M_G = (T_A \times 0.5) - (T_B \times 0.3)$

$I_G = Mk^2 =$

m = Composite mass of pulley

$= 500N = 500/9.81 = 50.96 kg$

$K = 0.4$



Sub α in

$I_G = 50.96 \times 0.4^2$

$\boxed{I_G = 8.155}$, $\alpha = a/r$ — tangential grade for composite pulley.

$$\boxed{(T_A \times 0.5) - (T_B \times 0.3) = -8.155 \times \alpha} \rightarrow (1)$$

sub T_A & T_B in (1)

$T_A = 101.936\,a + 1000$

$T_B = -203.87\,a + 2000$

Change $a \rightarrow \alpha$ $\quad a = \alpha r$

$r = 0.5$ for A block $\quad\quad r = 0.3$ for block B

T_A =

$a = \alpha \times 0.5$ $\quad\quad \boxed{a = 0.3\alpha}$

$\boxed{a = 0.5\alpha}$

sub a in T_B

sub in T_A

$T_A = 101.936(0.5\alpha) + 1000$ $\quad\quad T_B = -203.87(0.3\alpha) + 2000$

$\boxed{T_A = 50.968\alpha + 1000} \rightarrow (2)$ $\quad\quad \boxed{T_B = -61.161\alpha + 2000} \rightarrow (3)$

Sub. T_A & T_B in (1) eqn.

$(50.968\alpha + 1000)\,0.5 - [(-61.161\alpha + 2000)\,0.3] = -8.155\alpha$

$(25.484\alpha + 500) - [-18.3483\alpha + 600] = -8.155\alpha$

$[(25.484\alpha + 500) + (18.3483\alpha - 600)] = -8.155\alpha$

$43.832\alpha - 100 = -8.155\alpha$

$51.9873\alpha = 100$

$$\boxed{\alpha = 1.9235\ \mathrm{rad/s^2}}$$

MOMENTUM PRINCIPLE

Sub α in (2)

$T_A = 50.968(1.9235)+1000$

(a) $\boxed{T_A = 1098.03N}$

Sub α in (3)

$T_B = -61.161(1.9235)+2000$

$\boxed{T_B = 1882.35N}$

3. A homogeneous cylinder of weight 500N and radius 150mm is pulled by a horizontal force 200N through its mass centre, as shown in fig. Assuming the cylinder rolls without slipping, determine the angular acceleration of the cylinder. Take the coefficient of the friction of the contact surface as 0.25. Also calculate the distance moved by the cylinder in 2 seconds.

(b)

G.D

$W = 500N$

$r = 150mm.$

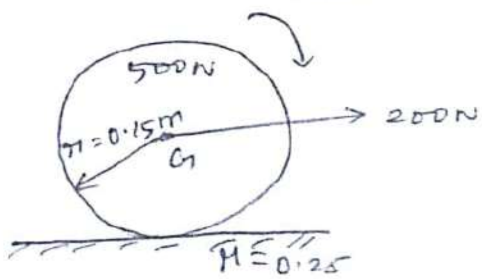
$F = 200N$

$\mu = 0.25$

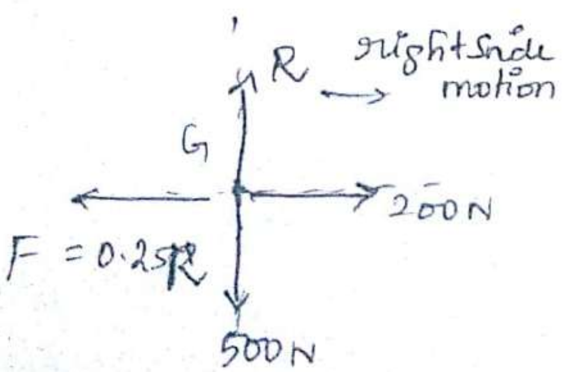
To find

$\alpha = ?$

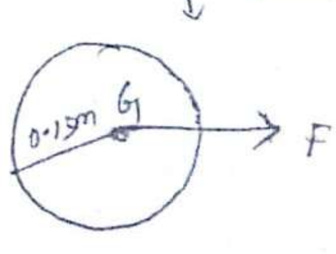
$S = ?$



FBD: block



FBD: pulley



$\Sigma F_x = +ma$

$\Sigma F_y = 0$

$\Sigma M_G = I_G \times \alpha$

$\Sigma M_G = F \times r$

$\boxed{\Sigma M_G = F \times 0.15}$

$, \alpha = a/r$

$\alpha = a/0.$

$\boxed{\alpha = 6.67}$

IMPULSE - MOMENTUM PRINCIPLE

For block: $\Sigma F \times t = m(v-u)$

$I(\omega - \omega_0)$

i.e.

5m

$\Sigma F_y = 0$

$R = 500N$

$\Sigma F_x = -0.25R + 200$

$= -0.25(500) + 200$

$\boxed{\Sigma F_x = 75}$, $m = 500/9.81$

$\Sigma F_x = ma$. $m = 50.96 kg$

$75 = 50.96 \times a$

$\boxed{a = 1.4715 m/s^2}$

$\alpha = \frac{1.4715}{0.15}$

$\boxed{\alpha = 9.81 rad/s^2}$

$I_G = \frac{mr^2}{2}$

$= \frac{50.96 \times 0.15^2}{2}$

$\boxed{I_G = 0.5733}$

$\Sigma M_G = I_G \times \alpha$

$(F \times 0.15) = (0.5733 \times 6.67a$

$F \times 0.15 = 3.823a$

$F = 25.49a$

$F = 25.49 \times 1.4715$

$\boxed{F = 37.5N}$

Scanned by CamScanner

IMPULSE - MOMENTUM PRINCIPLE

For block: $\Sigma F \times t = m(v-u)$

For pulley: $\Sigma M_G \times t = I_G(\omega - \omega_0)$

BLOCK

(a) If the block is moving in x-direction

$\Sigma F_x \times t = m(v-u)$, $\Sigma F_y = 0$

1. Right side motion

$\Sigma F_x \times t = m(v-u)$, $\Sigma F_y = 0$

2. Left side motion

$\Sigma F_x \times t = -m(v-u)$, $\Sigma F_y = 0$

(b) If the block is moving in y-direction

$\Sigma F_y \times t = m(v-u)$, $\Sigma F_x = 0$

1. Upward motion

$\Sigma F_y \times t = m(v-u)$, $\Sigma F_x = 0$

2. Downward motion

$\Sigma F_y \times t = -m(v-u)$, $\Sigma F_x = 0$

pulley

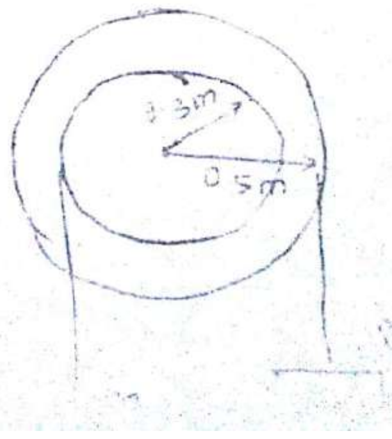
(a) If the pulley is rotating in clockwise direction

$\Sigma M_G \times t = I_G(\omega - \omega_0)$

(b) If the pulley is rotating in anticlockwise direction

$\Sigma M_G \times t = -I_G(\omega - \omega_0)$

1. A composite pulley of weight 500N and radius of gyration 0.4m is attached to 2 block of weight 1000N and 2000N resp a shown in fig. determine the velocity of block after 5 second.



I.P

$W_p = 500\,N$

$k_p = 0.4\,m$

$W_A = 1000N$

$W_B = 2000\,N$

$t = 5\,sec$

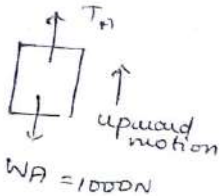
$u = 0$

To find

$v = ?$

Soln:

FBD: A block



$\Sigma F_y \times t = m(v-u)$

$W_A = 1000N$

$m = 1000/9.81$

$\boxed{m = 101.94\,kg}$

$\boxed{\Sigma F_y = T_A - 1000}$

Sub ΣF_y, m in main formula:

$(T_A - 1000) \times 5 = 101.94\,(v-0)$

$T_A - 1000 = 20.388v$

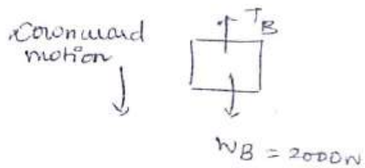
$\boxed{T_A = 20.388 v_A + 1000}$ → (1)

$v_A = \omega \times r = 0.5\omega$

$T_A = 20.388\,(0.5\omega) + 1000$

$\boxed{T_A = 10.194\omega + 1000}$ → (3)

FBD: B block



$\Sigma F_y \times t = -m(v-u)$

$W_B = 2000N$

$\boxed{m_B = 203.87\,kg}$

$\boxed{\Sigma F_y = T_B - 2000}$

"

$(T_B - 2000)\,5 = -203.87\,(v$

$T_B - 2000 = -40.774v$

$\boxed{T_B = -40.77 v_B + 2000}$ → (2)

$v_B = 0.3\omega$

$T_B = -40.77(0.3\omega) + 2000$

$\boxed{T_B = -12.231\omega + 2000}$ → (4)

FBD: Pulley

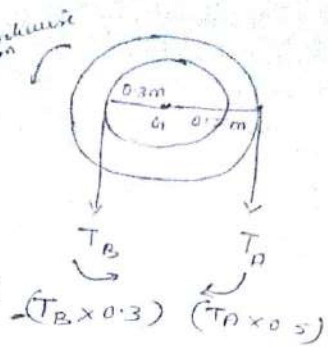
$k = 0.4m$, $W = 500N$

$I_G = Mk^2$

$= \frac{500}{9.81} \times 0.4^2$

$\boxed{I_G = 8.155}$

Anticlockwise rotation



$\Sigma M_G = (T_A \times 0.5) - (T_B \times 0.3)$

Main formula:

$$\Sigma M_G \times t = -I_G(\omega - \cancel{\omega_0}^{0})$$

$$(0.5T_A - 0.3T_B)5 = -8.155(\omega)$$

$$\boxed{0.5T_A - 0.3T_B = -1.631\omega} \longrightarrow (5)$$

Sub T_A (3) eqn & T_B (4) eqn in (5) eqn.

$$0.5(10.194\omega + 1000) - 0.3(-12.231\omega + 2000) = -1.631\omega$$

$$5.097\omega + 500 + 3.669\omega - 600 = -1.631\omega$$

$$8.766\omega - 100 = -1.631\omega$$

$$\boxed{\omega = 9.618 \text{ rad/s}}$$

$V_A = 0.5\omega$, $V_B = 0.3 \times \omega$

$V_A = 0.5 \times 9.618$

$V_B = 0.3 \times 9.618$

$\boxed{V_A = 4.809 \text{ m/s}}$

$\boxed{V_B = 2.8854 \text{ m/s}}$

Two blocks A and B of mass 250 kg and 200 kg are connected by a string, which passes over a frictionless disc of mass 50 kg as shown in fig. Calculate the velocity of the blocks after 5 seconds when they are released from rest.

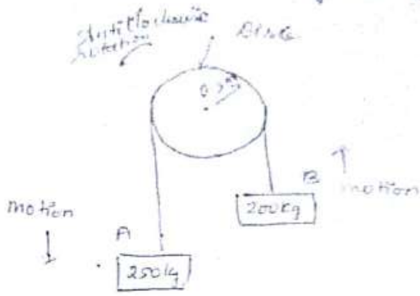
<u>G.P</u>

$m_A = 250kg$

$m_B = 200kg$, $u = 0$

$m_p = 50kg$,

$t = 5$ seconds



<u>To find</u>

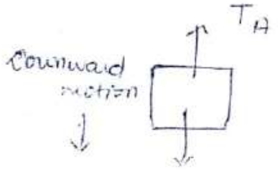
$v = ?$

<u>Soln</u>

<u>FBD : A block</u>

$m_A = 250kg$

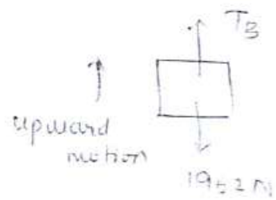
$W_A = 2452.5N$.



<u>FBD : B block</u>

$m_B = 200kg$.

$W_B = 1962N$



$\Sigma F_y \times t = -m(v-u)$

$\Sigma F_x = 0$ (Not used)

$\boxed{\Sigma F_y = T_A - 2452.5}$

$(T_A - 2452.5)\,5 = -250(v-0)$

$\boxed{T_A - 2452.5 = -50v}$, $v = 10.7$

$= 410.5$

$\Sigma F_y \times t = m(v-u)$

$\Sigma F_x = 0$ (Not used)

$\boxed{\Sigma F_y = T_B - 1962}$

$$2452.5 = -25\omega$$

$$(T_B - 1962) = 40(v - 0)$$

$$\boxed{T_B - 1962 = 40v} \quad v = 0.5 \times \omega$$

$$T_B - 1962 = 40(0.5 \times \omega)$$

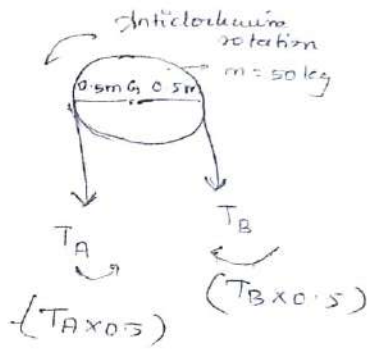
$$\boxed{T_B - 1962 = 20\omega}$$

$$\boxed{T_B = 20\omega + 1962} \rightarrow (2)$$

$$\boxed{T_A - 2452.5 = -25\omega}$$

$$\boxed{T_A = -25\omega + 2452.5} \rightarrow (1)$$

FBD; pulley



$m_P = 50kg$, $\omega_0 = 0$

$W_P = 50 \times 9.81 = 490.5 N$

$$\Sigma M_G \times t = -I_G(\omega - \omega_0)$$

$$\Sigma M_G = T_B \times 0.5 - T_A \times 0.5$$

$$\boxed{\Sigma M_G = 0.5T_B - 0.5T_A}$$

$$I_G = \frac{mr^2}{2} = \frac{50 \times 0.5^2}{2} = 6.25 \ kg\,m^2$$

$$(0.5T_B - 0.5T_A)\,5 = -6.25\omega$$

$$\boxed{0.5T_B - 0.5T_A = -1.25\omega} \rightarrow (3)$$

Sub (1) & (2) in (3) equation.

$$0.5(20\omega + 1962) - 0.5(-25\omega + 2452.5) = -1.25\omega$$

$$10\omega + 981 + 12.5\omega - 1226.25 = -1.25\omega$$

WORK - ENERGY PRINCIPLE
[motion in y-direction]
+ x-direction

Block pulley

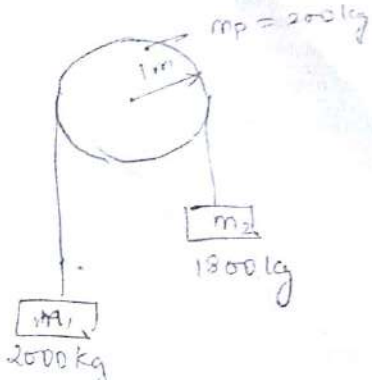
" ($\leq F \times e$)

1. An elevator system is mounted ... $m_1 = 2000$ kg, $m_2 = 1800$ kg and a uniform disk of $m_3 = 200$ kg. Calculate the velocity of m_1 at time $t = 10$ sec as system starts from rest. (Using impulse momentum method)

Ans: $T_A = 18613.84$ N

$T_B = 18563.53$ N

$\omega = 5.03$ rad/s



Work - Energy Principle:

Work - Energy Principle

[motion in y-direction]
[in x-direction]

Block

1. $\left(\Sigma F_y \times s\right) = \frac{m}{2}(v^2 - u^2)$ — upward motion

$\Sigma F_x = 0$ (not used)

2. Block moving in downward motion

$\Sigma F_y \times s = \frac{-m}{2}(v^2 - u^2)$

$\Sigma F_x = 0$ (not used)

pulley

$(\Sigma M_G \times \theta) = \frac{I_G}{2}(\omega^2 - \omega_0^2)$

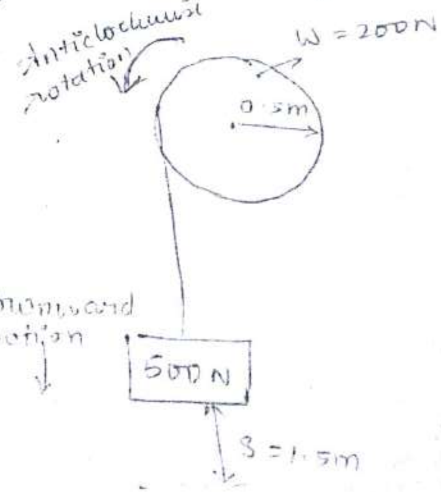
1. If the pulley is rotating in clockwise direction

$(\Sigma M_G \times \theta) = \frac{I_G}{2}(\omega^2 - \omega_0^2)$

2. If the pulley is rotating in Anticlockwise direction

$(\Sigma M_G \times \theta) = -\frac{I_G}{2}(\omega^2 - \omega_0^2)$

① A block of 500N is suspended by a tie rope around the pulley of weight 200N and radius 0.5m as shown in fig. Determine the velocity of the weight and tension in the rope after moving a distance of 1.5m



G.D

$W_B = 500N$

$m_B = \frac{500}{9.81} = 50.968\,kg$

$W_P = 200N$

$m_P = \frac{200}{9.81} = 20.38\,kg$

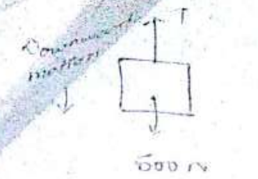
$r = 0.5m$

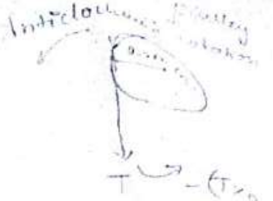
$S = 1.5m$, $u = 0$ (rest)

To find:

$v^2 = ?$

$T = ?$





$\Sigma F_y \times S = \frac{-m}{2}(V^2 - u^2)$

$\Sigma F_x = 0$ (Not used)

$\Sigma F_y = T - 500$

$(T - 500) \times 1.5 = \frac{-50.968}{2} V^2$

$T - 500 = -16.98 V^2$

$$\boxed{T + 16.98 V^2 = 500} \rightarrow ①$$

$\Sigma M_G \times \theta = \frac{-I_G}{2}(\omega^2 - \omega_0^2)$

$$\boxed{\Sigma M_G = -T \times 0.5}$$

$I_G = \frac{m r^2}{2}$

$= \frac{20.38 \times 0.5^2}{2}$

$$\boxed{I_G = 2.5475 \text{ kg}\cdot\text{m}^2}$$

$\theta = S/r, \; \omega = V/r$

$(-T \times 0.5) \times (S/r) = \frac{-2.5475}{2}(V/r)^2$

$(-T \times 0.5) \times \left(\frac{1.5}{0.5}\right) = \frac{-2.5475}{2}\left(\frac{V^2}{0.5^2}\right)$

$1.5T = 5.095 V^2$

$$\boxed{T = 3.396 V^2} \rightarrow ②$$

Sub $T = 3.396V^2$ in ①

$3.396V^2 + 16.98V^2 = 500$

$$\boxed{V = 4.4533 \text{ m/s}}$$

$$\boxed{\omega = V/r = \frac{4.4533}{0.5} = 8.903 \text{ rad/s}}$$

A drum of mass 20kg shown in fig is subjected to a moment of 160 N m. Determine the speed of the 8 kg block B if it rises 1.30m starting from rest. Use Work Energy principle. The radius of gyration of the drum about 'O' is 450mm

G.D

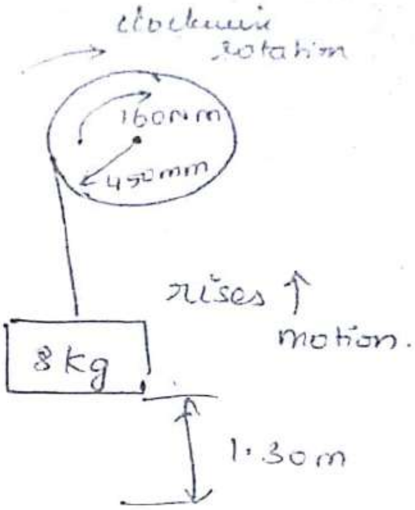
$m_D = 20\,kg$

$m_b = 8kg$

rest $u = 0, \omega_0 = 0$

$K = 450mm = 0.45m$

$s = 1.30m.$



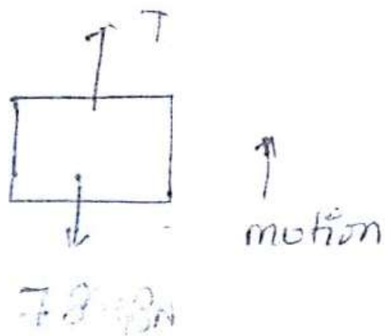
To find

$v = ?$

Block

$m_b = 8kg$

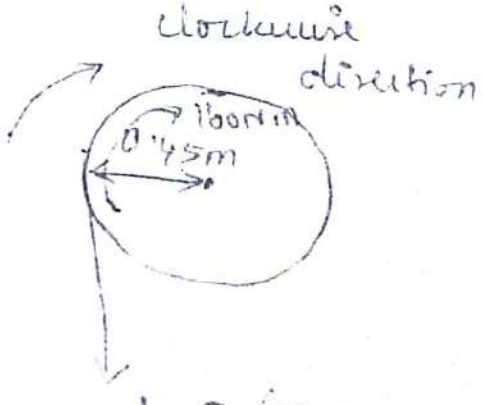
$W_b = 78.48 N$



Pulley (Drum)

$m_D = 20\,kg$

$W_D = 196.2\ N$



$\Sigma F_y \times s = \frac{m}{2}(v^2 - u^2)$

$T \curvearrowright (T \times 0.45)$

$\Sigma F_x = 0$ (not used)

$I_G = mk^2$

$= 20 \times 0.45^2$

$\boxed{\Sigma F_y = T - 78.48 \text{ ↑}}$

$\boxed{I_G = 4.05\ kg \cdot m^2}$

$$\times S = \frac{m}{2}(v^2 - u^2) \qquad \Sigma M_G \times \theta = \frac{I_G}{2}(\omega^2 - \omega_0^2)$$

$$(T - 78.48)\,1.5 = \frac{8}{2} v^2 \qquad 160 - T \times 0.45) \times 2.87$$

$$T - 78.48 = 3.076 v^2 \qquad \omega = V/r = V/0.45 \quad \frac{4.05}{2}(\omega^2)$$

$$\boxed{T - \underbrace{3.076}_{b_1} v^2 = \underbrace{78.48}_{c_1}} \rightarrow (1) \qquad 160 - T \times 0.45) = \frac{4.05}{2 \times 2.87}\left(\frac{v^2}{0.2025}\right)$$

$$160 - 0.45T = 3.46 v^2$$

$$-0.45T - 3.46 v^2 = -160$$

$$\underbrace{0.45}_{a_2} T + \underbrace{3.46}_{b_2} v^2 = \underbrace{160}_{c_2}$$

$$\boxed{T = 157.65 N}$$

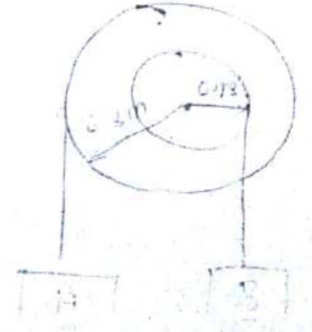
$$V^2 = 25.73$$

$$\boxed{V = 5.072 m/s}$$

Exercise

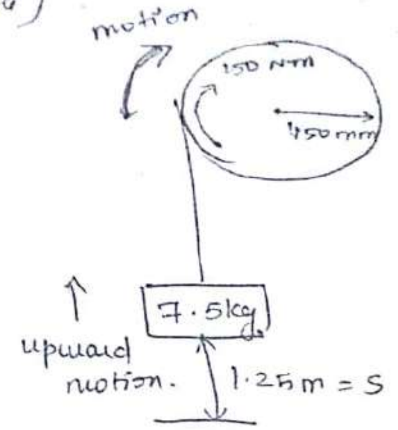
1. Two blocks A and B of mass 25kg and 10 kg are suspended by compound pulley of mass 30kg and radius of gyration 0.25m as shown in fig. If the blocks are released from rest, determine time required for block A to attain a speed of 2m/s. (Impulse momentum principle)

Ans. $\boxed{t = 0.53 \text{ sec}}$



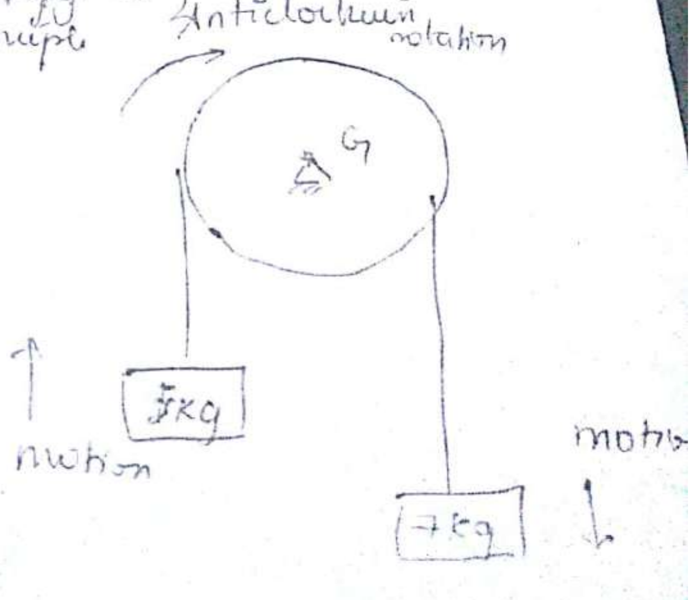
2: The drum of mass 15kg shown in fig. has a radius of gyration about 'O' of 250mm. determine the speed of 7.5kg block B after it rises 1.25m starting from rest. The drum is also subjected to a moment of 150NM as shown.

(Work - Energy Principle)



3: A massless rope carries two masses of 5kg and 7kg when hanging on a pulley of mass 5kg, radius 600mm, and radius of gyration 450mm. At what distance, it will take to change the speed of the masses from 3m/s to 6m/s

(Work - Energy principle)



G.D

$m_A = 5kg$

$m_B = 7kg$

$u = 3m/s,$

$V = 6m/s$

$r = 600mm$

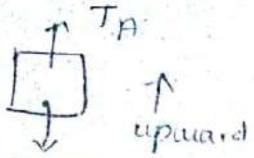
$r = 0.6m$

$k = 0.45m$

To find

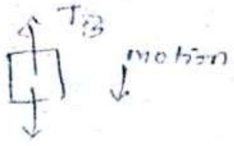
Block

A



5×9.81
$= 49.05N$

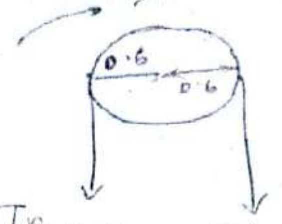
B

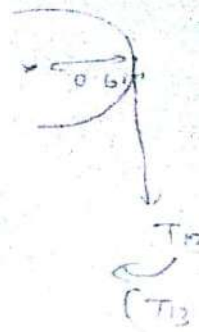


7×9.81
$= 68.67N$

pulley

Anticlockwise





$-(T_A\times 0.6)$ $(T_B\times 0.6)$

A - block

$\Sigma F_y \times s = \frac{m}{2}(v^2-u^2)$

$\Sigma F_x \neq 0$ n.u

$\boxed{\Sigma F_y = T_A - 49.05}$

B - block

$\Sigma F_y \times s = -\frac{m}{2}(v^2-u^2)$

$\boxed{\Sigma F_y = T_B - 68.67N}$

$(T_A - 49.05)\,s = \frac{5}{2}(6^2-3^2)$ $(T_B - 68.67)\,s = -\frac{7}{2}(6^2-3^2)$ 10

$\boxed{sT_A - 49.05s = 67.5}$ (1) $\boxed{sT_B - 68.67s = -94.5}$ (2)

$\underset{a_1}{1}\,sT \quad \underset{b_1}{-49.05}\,s = \underset{c_1}{67.5}$

$\underset{a_2}{1}\,sT \quad \underset{b_2}{-68.67}\,s = \underset{c_2}{-94.5}$

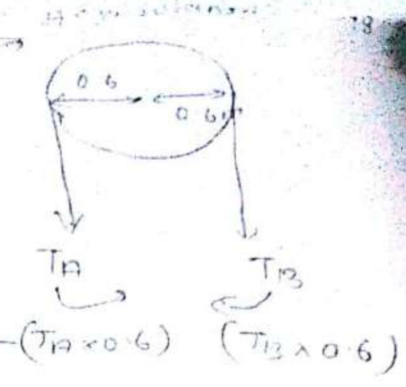
$sT = 472.5$

$\therefore s = 8.256\ m$

$T = \frac{472.5}{8.256} =$

Pulley

$\Sigma M_G \times S = \frac{-I_G}{2}(\omega^2 - \omega_0^2)$



$\omega = V/r$

$\omega = V/0.6 = 6/0.6 = 10$ rad/s

$\omega_0 = u/0.6 = 3/0.6 = 5$ rad/s

$\Sigma M_G = (T_B \times 0.6) - (0.6\, T_A)$

$k = 450$ mm $= 0.45$ m

$I_G = mk^2$

$= 5 \times 0.45^2$

$\boxed{I_G = 1.0125 \text{ kg m}^2}$

$\left(T_B \times 0.6 - 0.6\, T_A\right) S = -\frac{1.0125}{2}(10^2 - 5^2)$

$(0.6\, T_B - 0.6\, T_A)\, s = -37.96 \rightarrow (3)$

(1) & (2)

$T_A = \frac{67.5}{s} + 49.05$

$T_B = -\frac{94.5}{s} + 68.67$

sub (1) & (2) in (3)

$\left(0.6\left(-\frac{94.5}{s} + 68.67\right) - 0.6\left(\frac{67.5}{s} + 49.05\right)\right) s$

$= -37.96$

$\left(-\frac{56.7}{s} + 41.202 - \frac{40.5}{s} - 29.43\right) s = 37.96$

$-\,97.2 \;-\; 11.772\,S = -37.96$

$-\,11.772\,S = 59.24$

$$\boxed{S = 5.03\,m}$$

* 2 mark

Instantaneous centre of rotation (IC)

When a body is in general plane motion, it undergoes both translation and rotation. For the sake of simplicity, this combined motion of translation and rotation, may be assumed to be a motion of pure rotation about a centre point.